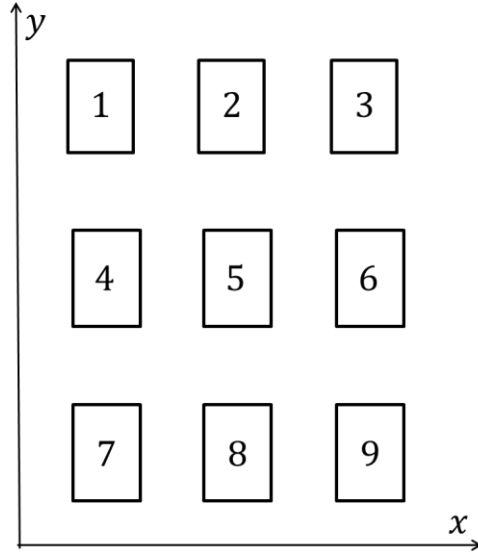


## Control of a vehicle formation

Consider the set of nine vehicles depicted in the following figure.



Each vehicle represents a “subsystem” of the formation and is characterized by four states:

- $d_i^x$ : displacement with respect to the desired distance between vehicle  $i$  and the vehicle on its left (in case of vehicle 1, it is its displacement with respect to the rest desired position; in case of vehicles 4 and 7, it is the difference of their position with respect to the one of the predecessor vehicle), **in the x direction**;
- $v_i^x$ : velocity of vehicle **in the direction x**;
- $d_i^y$ : displacement with respect to the desired distance between vehicle  $i$  and its predecessor (in case of vehicle 1, it is its displacement with respect to the rest desired position; in case of vehicles 2 and 3, it is the difference of their position with respect to the one of the vehicle on the left) **in the direction y**;
- $v_i^y$ : velocity of vehicle  $i$  **in the direction y**.

The input vector  $u_i = [u_i^x, u_i^y]'$  is the force exerted on the vehicle in horizontal and vertical directions, respectively and a friction coefficient  $h_i$  is defined for each vehicle. For vehicle  $i = 1$ , the dynamics is

$$\begin{cases} \dot{d}_i^x = v_i^x \\ \dot{v}_i^x = -\frac{h_i}{m_i} v_i^x + \frac{1}{m_i} u_i^x \\ \dot{d}_i^y = v_i^y \\ \dot{v}_i^y = -\frac{h_i}{m_i} v_i^y + \frac{1}{m_i} u_i^y \end{cases}$$

For  $i=4,7$ :

$$\begin{cases} \dot{d}_i^x = v_i^x - v_{i-3}^x \\ \dot{v}_i^x = -\frac{h_i}{m_i} v_i^x + \frac{1}{m_i} u_i^x \\ \dot{d}_i^y = v_i^y - v_{i-3}^y \\ \dot{v}_i^y = -\frac{h_i}{m_i} v_i^y + \frac{1}{m_i} u_i^y \end{cases}$$

For  $i=2,3$ :

$$\begin{cases} \dot{d}_i^x = v_i^x - v_{i-1}^x \\ \dot{v}_i^x = -\frac{h_i}{m_i} v_i^x + \frac{1}{m_i} u_i^x \\ \dot{d}_i^y = v_i^y - v_{i-1}^y \\ \dot{v}_i^y = -\frac{h_i}{m_i} v_i^y + \frac{1}{m_i} u_i^y \end{cases}$$

For  $i=5,6,8,9$ :

$$\begin{cases} \dot{d}_i^x = v_i^x - v_{i-1}^x \\ \dot{v}_i^x = -\frac{h_i}{m_i} v_i^x + \frac{1}{m_i} u_i^x \\ \dot{d}_i^y = v_i^y - v_{i-3}^y \\ \dot{v}_i^y = -\frac{h_i}{m_i} v_i^y + \frac{1}{m_i} u_i^y \end{cases}$$

Overall, the system dynamics can be described by the following model:

$$\dot{x} = Ax + Bu$$

where  $x = [x_1^T, \dots, x_9^T]^T$  (where  $x_i = [d_i^x, v_i^x, d_i^y, v_i^y]^T$  for all  $i = 1, \dots, 9$ ) and  $u = [u_1^T, \dots, u_9^T]^T$ , respectively. The matrices are specified in the corresponding MATLAB file. Note that all state variables are measurable.

### Problem:

1. Decompose the state and input vectors into subvectors, consistently with the physical description of the system. Obtain the corresponding decomposed model.
2. Generate the system matrices (both continuous-time and discrete-time, the latter with a sampling time  $h = 0.1$  s). Perform the following analysis:
  - a. Compute the eigenvalues and the spectral abscissa of the (continuous-time) system. Is it open-loop asymptotically stable?
  - b. Compute the eigenvalues and the spectral radius of the (discrete-time) system. Is it open-loop asymptotically stable?
3. For different state-feedback control structures (i.e., centralized, decentralized, and different distributed schemes) perform the following actions
  - a. Compute the continuous-time fixed modes
  - b. Compute the discrete-time fixed modes

- c. Compute, if possible, the CONTINUOUS-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
- d. Compute, if possible, the DISCRETE-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
- e. Analyze the properties of the so-obtained closed-loop systems (e.g., stability, eigenvalues) and compute the closed-loop system trajectories (generated both in continuous-time and in discrete-time) of the distances between the vehicles starting from a common random initial condition.